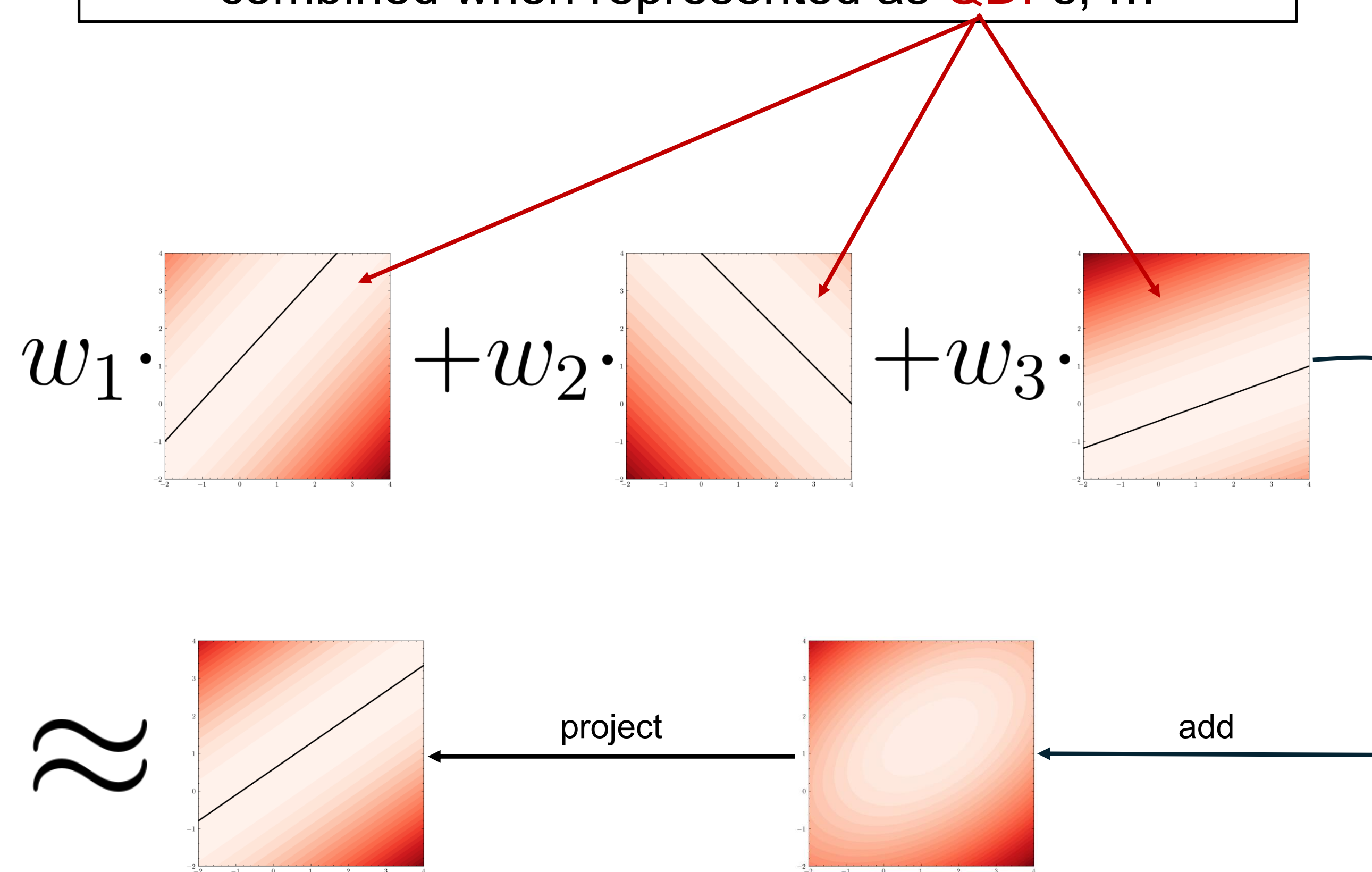
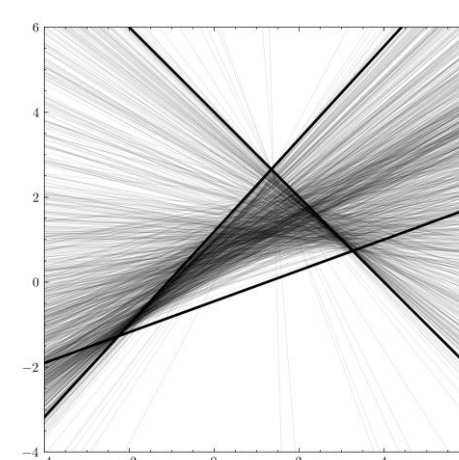


Background: Affine spaces (flats) can be linearly combined when represented as **QDFs**, ...



resulting in a new affine subspace.

Convex hull of the
basis lines:



But: Given a basis of flats and a target flat.

How to compute the coefficients?

Method: Solve a small linear system!

$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$\rightarrow k(d-k)$ linear equations for orientation
 $+ d-k$ equations for displacement
 $+ 1$ equation for sum-to-one property
 $= (k+1)(d-k)+1$ equations in total

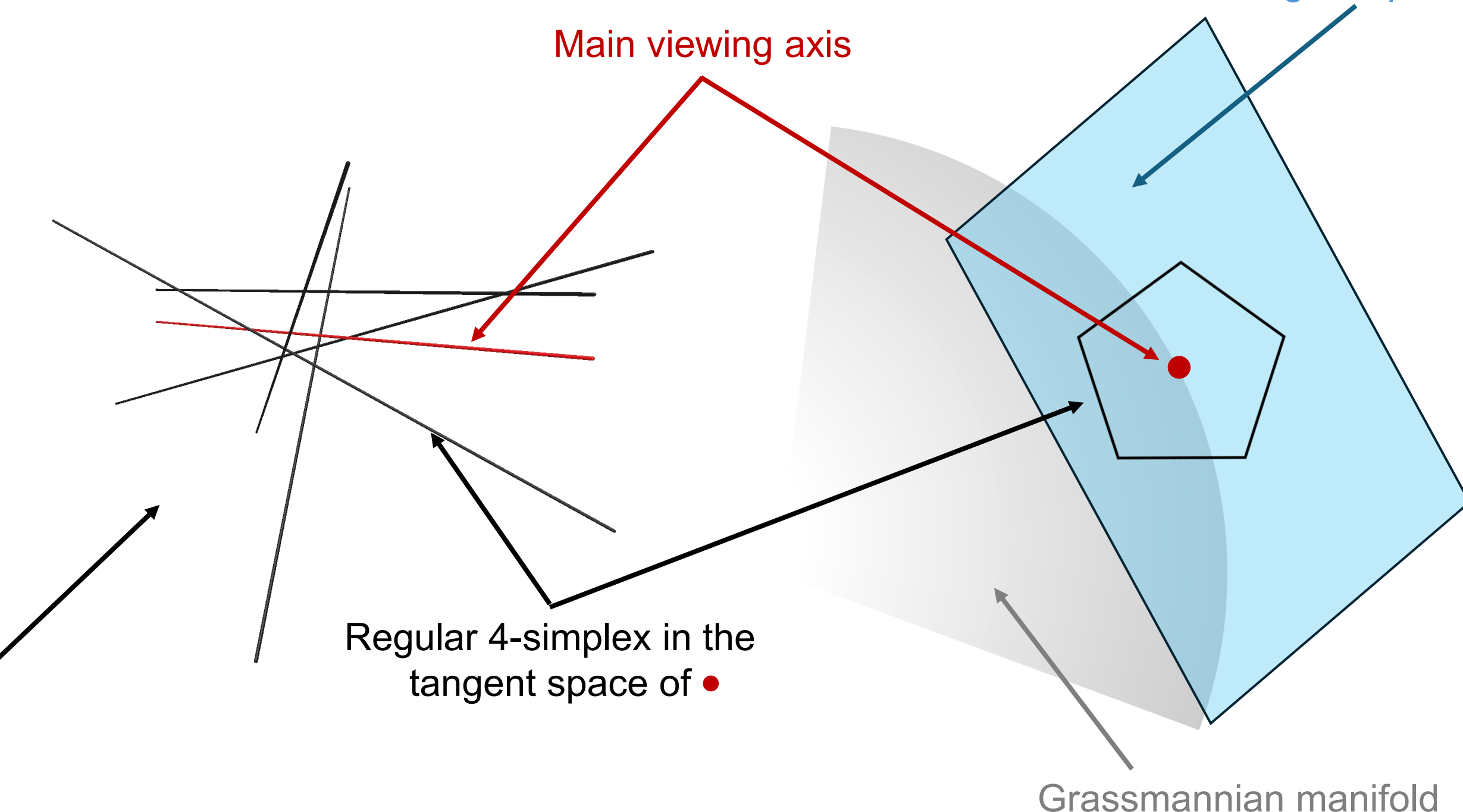
In practice:

Lines in 3D: 5 equations
 Planes in 3D: 4 equations

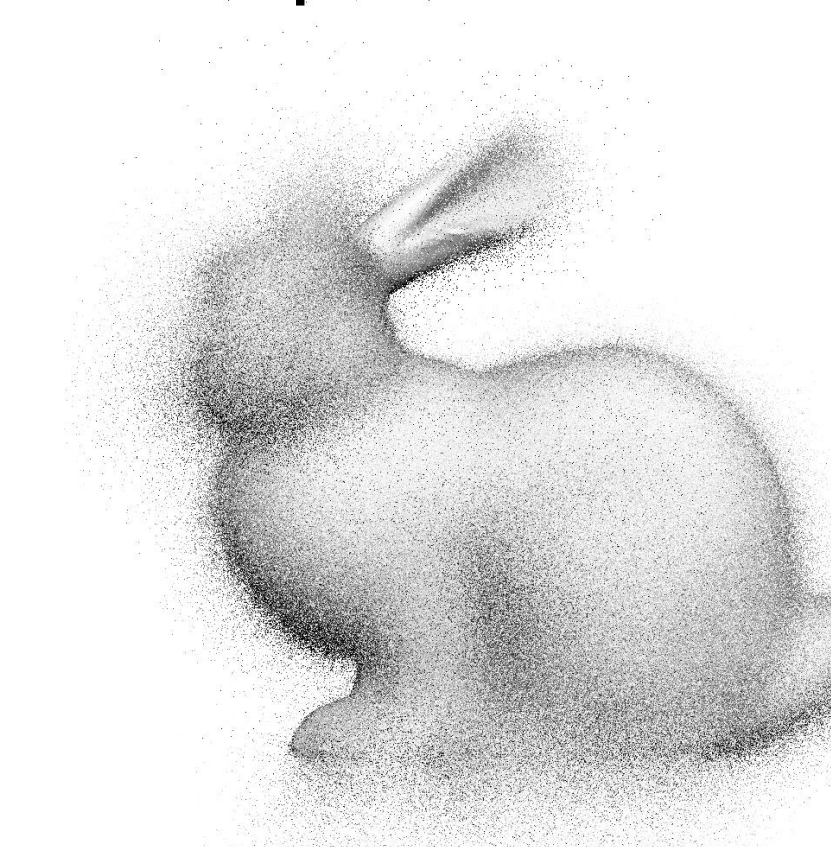
Application: Scattered data interpolation
via triangulations of coordinates

Here: View interpolation

Represent relevant lines w.r.t. a base simplex:



Input:



Interpolation results:

